TALENT.AI LAB · GRADUATION RESEARCH TOPICS

Research Frontiers in Applied AI for Financial Services

Six research topics you can take on as a graduation project, co-supervised by Finaira.

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Finaira · Talent.AI Lab	**University Students**	**Graduation Project Topics**

ABOUT THIS BRIEF

Finaira builds AI systems for banks and financial institutions across the MENA region. Banking is a demanding place to put AI to work: systems often have to run on the bank's own servers and infrastructure rather than on the cloud, sensitive customer data cannot leave their premises, and everything has to run efficiently on, at times, very limited hardware. Those constraints rule out simply reaching for the biggest available cloud model, which is what makes the research genuinely open.

Below are six research topics in that space, any of which you can take on as a graduation project with Finaira co-supervising alongside your university department. Each topic lists example applications, plus the gaps where a contribution is still open. The first three are defined starting points with a fairly clear problem to begin from. The last three are open-ended areas where you help shape the specific question.

PART 1 · DEFINED STARTING POINTS

TOPIC 1 · RETRIEVAL

Efficient search over large, cross-referencing document collections

To answer questions about a large pile of documents, an AI system usually searches for the relevant passages first, then hands only those to a language model to phrase the answer. The quality of the final answer depends heavily on that search step: retrieve the wrong passages and even the best model answers wrongly. Banking collections make this hard, mixing Word, Excel, and PDF sources across many topics, with documents referring to one another and a single subject appearing on page 10 for example of a file and again, differently worded, on page 90. Information sparsity is the topic in question here. The open question is how good that search step can get, through smarter ways of splitting documents into

pieces, indexing them, and matching them to a question, so the system stays fast and light enough to run on-site instead of leaning on an expensive large model for every query.

EXAMPLE APPLICATIONS

- Question answering over regulatory, credit, and policy document sets, served efficiently on-site.
- Long documents where the relevant information is scattered across distant pages.
- Search over collections that mix Arabic and English, where question and source may be in different languages.

WHERE A CONTRIBUTION IS OPEN

- Retrieval benchmarks are mostly clean, single-document English text, not messy multi-format financial collections.
- Chunking and indexing strategies are rarely compared under a fixed compute budget.
- Arabic-English retrieval, where a question in one language must find its answer in the other, has little published work.
- RAG optimizations at different stages.
- How much work a good index can take off the language model is still unsettled.

TOPIC 2 · PRIVACY

Hiding sensitive data so it can safely leave the building, then putting it back

Some tasks would benefit from a powerful cloud model, but regulation often forbids sensitive data from leaving the company premises. One promising idea is to transform a document before it leaves: strip out or disguise details such as names and account numbers, send the safe version off to be processed, then map the results back onto the original. Making that work is the research problem. The transformation has to be reversible, it has to hold up on financial text mixing Arabic and English, and it has to keep the document searchable and understandable even after the sensitive parts are hidden. Detecting sensitive information in Arabic is also noticeably harder than in English, and is an open problem in its own right.

EXAMPLE APPLICATIONS

- Letting a bank use stronger off-site models for selected tasks without exposing customer data.
- Detecting names, account numbers, and other sensitive entities in Arabic financial text.
- Deciding what may safely leave the premises under regional data-protection rules.

WHERE A CONTRIBUTION IS OPEN

- Labelled Arabic datasets for sensitive-entity detection are scarce, and detection quality lags English.
- The effect of anonymization and/or encryption on retrieval and answer quality is rarely measured.
- Keeping the disguising consistent and reversible across a whole corpus, not one document at a time.
- No settled way to connect a technical anonymization scheme to what regional regulation requires.

TOPIC 3 · GUARDRAILS

Small, bilingual safety checks that run on-site

Guardrails are the safety components that sit around an AI system and check what goes in and what comes out, watching for unsafe content, policy violations, leaked personal data, and attempts to trick the system into misbehaving. Good tools exist, but most assume a large model on serious hardware, and many work poorly on Arabic or on mixed-language text. There is also a known weakness: a guardrail that inspects only a short window of text can sometimes be slipped past using a long or broken-up input. The opportunity is to build guardrails light enough to run efficiently in production, effective across both Arabic and English, and robust against those evasion tricks. Recent evidence that surprisingly small models can approach much larger ones on safety tasks suggests this is a realistic target.

EXAMPLE APPLICATIONS

- Always-on safety and policy checking that runs efficiently inside on-site deployments.
- Detecting sensitive-data leakage in model inputs and outputs, in both Arabic and English.
- Enforcing the data-movement rules from Topic 2 at the point where data would leave.

WHERE A CONTRIBUTION IS OPEN

- Safety evaluation sets are overwhelmingly English; Arabic and language-switched test sets barely exist.
- Evasion through long or fragmented inputs is documented, but defences against it are thin.
- How far a guardrail can shrink before its robustness collapses has not been mapped.
- Guardrails are studied as content filters, rarely as enforcers of data-movement policy.
- Guardrailing modularity.

PART 2 · OPEN-ENDED DIRECTIONS

TOPIC 4 · SIMULATION

Simulating markets and customers with populations of AI agents

Generative AI can be used to build a simulator: populate a virtual world with many AI agents standing in for market participants or bank customers, then watch how they behave. Built once, it becomes a platform for a whole line of study, letting you observe patterns emerge, stress-test other models against unusual conditions, and try interventions before touching a real system. The open questions are unsettled, which is what makes this an area to explore. Whether AI agents behave like believable economic actors is contested, and deciding when a simulation is realistic enough to trust is itself among the hardest parts. A separate line uses these models to manufacture synthetic customer and transaction data, where the question is whether what you learn from the synthetic data still holds on real data.

EXAMPLE APPLICATIONS

- Stress-testing predictive and decision-making systems against simulated market and customer conditions.
- Generating synthetic customer and transaction data when real data cannot be used.
- Modeling financial-inclusion scenarios, such as how underserved segments respond to a new product.

WHERE A CONTRIBUTION IS OPEN

- No agreed way to decide when a simulated population is realistic enough to trust. This is the field's central open problem.
- Studies disagree on whether AI agents behave like believable economic actors.
- Whether findings from synthetic data hold on real data is more often asserted than tested.
- Calibration to GCC or Egyptian markets and Arabic-speaking populations is almost unexplored.

TOPIC 5 · ADVISORY

AI-driven investment advisory: risk, forecasting, and strategy

An automated advisory pipeline runs the whole chain from raw market data to investment guidance: gauging how risky the market is, forecasting how prices might move, then recommending stocks, allocations, and strategies matched to an investor's appetite for risk. Each link is its own research surface. Risk measurement brings in volatility and tail-risk estimation, forecasting draws on deep-learning and transformer models for financial time series, and recommendation connects to portfolio optimization and, increasingly, to language-model-based financial analysis. The field is moving from simple one-shot allocation toward systems that adjust continuously, and how much of the decision to automate remains open.

EXAMPLE APPLICATIONS

- Measuring and forecasting market risk for regional stocks and portfolios.
- Recommendation engines tuned to an investor's risk profile and to local market structure.
- Extracting usable market signals from Arabic-language news and reports.

WHERE A CONTRIBUTION IS OPEN

- Most published work is built and tested on US and European market data, not regional markets.
- Extracting market signals from Arabic-language news and reports is largely unstudied.
- The move from single-period to continuous, multi-period allocation is active and unsettled.
- How to earn an investor's trust, and how much to automate, remains an open design question.
- These pipelines assume cloud-scale compute; behaviour under on-site constraints is untested.

TOPIC 6 · SMALL MODELS

Small language models: how much can you do with a little?

Small language models, from a few hundred million up to a few billion parameters, are the thread running under much of this brief, since several topics above depend on models compact enough to run efficiently on-site. As a research area in their own right they raise a clear set of questions: how much capability fits into a small model, how you specialize one for a domain like banking, and how techniques such as distillation and quantization trade size against quality. There is real momentum behind the idea that small specialized models, rather than giant general ones, are the right tool for most repetitive and agent-style tasks.

EXAMPLE APPLICATIONS

- Running capable models entirely on-site, so data never has to leave the bank.
- Specializing a general model for banking and finance through fine-tuning and distillation.
- Compressing a model to fit the hardware a bank actually has available.

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WHERE A CONTRIBUTION IS OPEN

- Arabic and bilingual small models for finance have very little coverage and clear regional value.
- How small a model can be while staying useful for a given banking task is unmapped.
- Distillation, fine-tuning, and quantization are rarely compared head to head for finance.
- Benchmarks for small models on realistic banking tasks are largely missing.

AN INVITATION

All six topics are live, unsolved, and tied to real problems in the region's financial sector. Pick the one you would genuinely enjoy spending a year on, and Finaira will help supervise it into a strong graduation project.